Angio OMEX Combination Report — example_angio

example_angio | 1 replicate(s) | sim_time 100.0 MCS

Combination Overview

Property	Value
Scan point	example_angio
Replicates	1 successful (plotted and averaged)
Signals	num_clusters, cell_density, largest_cluster_size, vegf_mean
Time	MCS (angio has no physical-time calibration)

Provenance (OMEX Metadata)

Field	Value
Created	2026-10-19T21:03:18
Modified	2026-10-19T21:03:22.715527
License	https://spdx.org/licenses/CC-BY-4.0.html
Citation	Merks, R.M.H., Brodsky, S.V., Goligorksy, M.S., Newman, S.A., Glazier, J.A. (2006). Cell elongation is key to in silico replication of in vitro vasculogenesis and subsequent remodeling. Developmental Biology 289(1), 44-54. https://doi.org/10.1016/j.ydbio.2005.10.003
Keywords	angiogenesis, vasculogenesis, cellular Potts model, VEGF, agent-based model

Run Identity (stochastic realization & code)

Replicate	RNG seed	Origin	Schema fingerprint	Code
example_angio	799091067	drawn	sha256:0a6c2ceaca8b9cdf	built 043fb10

Replicates within a combination share a parameterization and differ only in the random number generator seed, so a difference between them is stochastic rather than configured. Seeds marked *drawn* were generated independently at launch and recorded afterwards; the software does not prescribe a seed. The schema fingerprint identifies the parameter definitions in force when the run was configured, and the code entry the source revision that executed it.

Per-replicate Archive Validation

Replicate	built-in SED-ML	manifest	biosimulators
example_angio	PASS	PASS	PASS (0 errors, 1 warning group)

Archive Contents (per replicate)

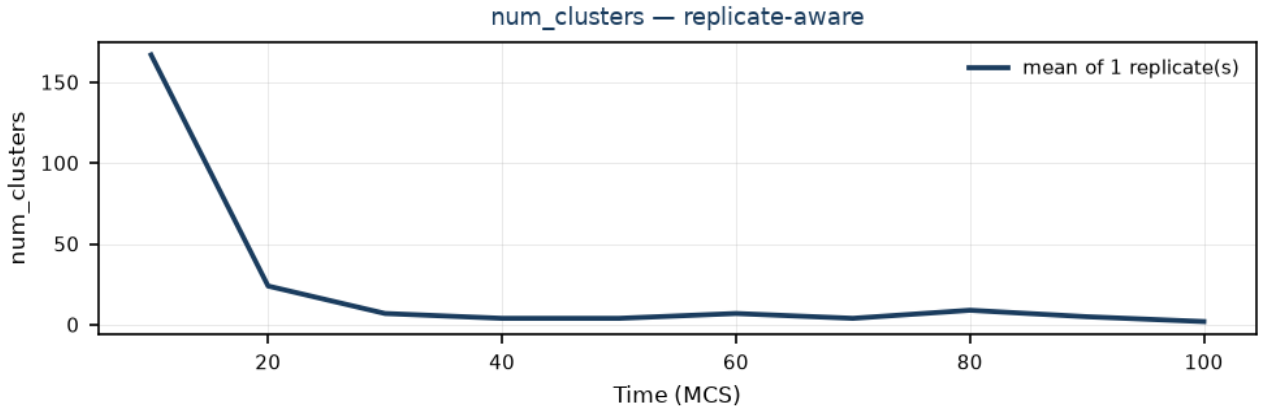
File	Size (B)	Role
logs/stderr.log	0	—
logs/stdout.log	1,803	—
manifest.xml	9,516	OMEX manifest (COMBINE archive index)
metadata.rdf	1,546	OMEX Metadata (provenance: creator, license, citation)
metrics.csv	353	Scalar signals over time (indexed by the SED-ML)
model/Simulation/angiogenesis.py	197	Bundled model file
model/Simulation/angiogenesis.xml	2,658	Bundled model file
model/Simulation/angiogenesisSteppables.py	1,260	Bundled model file
model/angiogenesis.cc3d	270	Bundled model file
model/environment.json	1,082	Bundled model file
model/model_summary.json	4,190	Bundled model artifact (model/@source resolves here)
model/parameter_audit.json	44,231	Bundled model file
model/run_model.py	789	Bundled model file
model/run_params.json	319	Bundled model file
model/src/angiogenesis_composer.py	1,116	Bundled model file

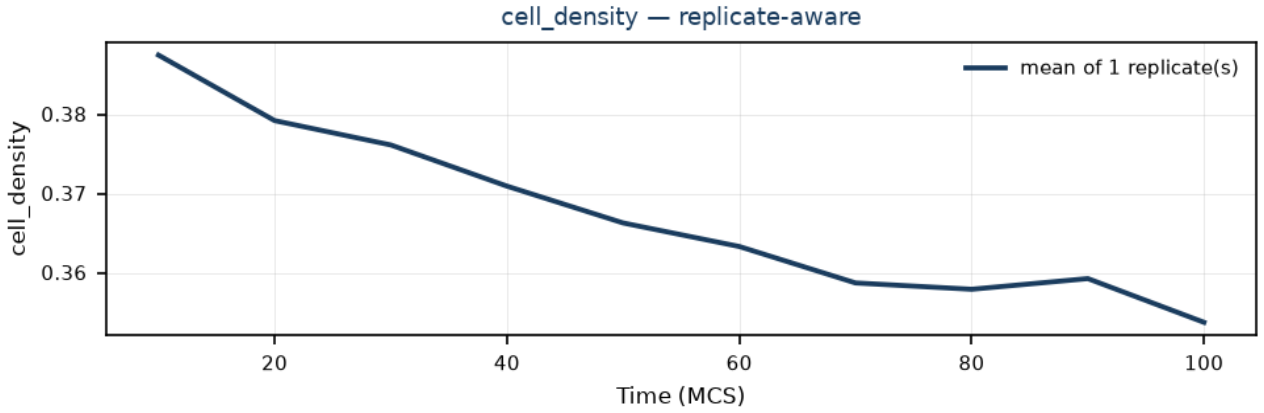
model/src/angiogenesis_process.py	99,294	Bundled model file
run_metadata.json	2,384	Run provenance / parameters
sim_params.json	319	Exact simulation parameters
simulation.sedml	5,391	SED-ML L1V4 simulation description
status.json	510	—
zarr/… (61 files)	2,064,209	Full Zarr spatial trajectory (per-timestep ndarrays)

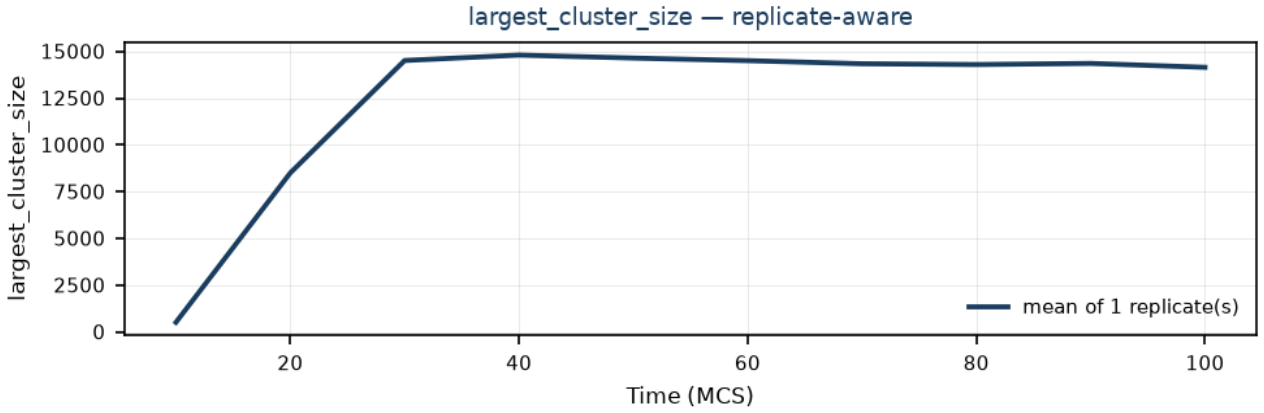
Each of the 1 replicate archives has this layout.

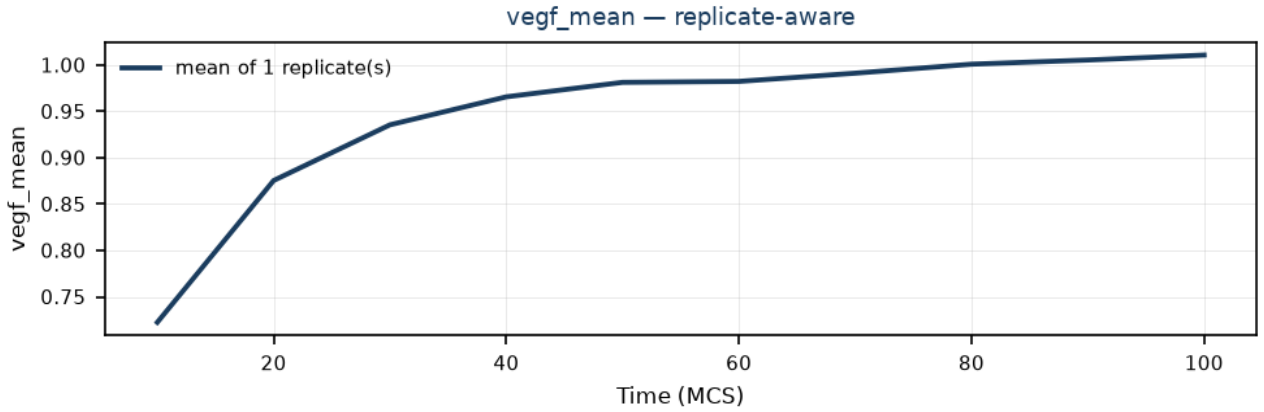
Metrics (replicate-aware, from the Zarr trajectory)

Thin = each replicate; bold = across-replicate mean. Time in MCS.









Final-step spatial fields (replicate 1)

